THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 9: **Symmetry** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** it can be folded along a line so that the two halves match	definition
2	**(b)** 3	from vertex to opposite side
3	**(a)** 6	same as the number of sides
4	**(d)** reflection	mirror image
5	**(c)** 4	fits 4 times in a turn
6	**(b)** 60°	360 ÷ 6
7	**(c)** rotational symmetry of order 2	half turn
8	**(c)** 2	both diagonals
9	**(a)** d	left-right swap
10	**(b)** Parallelogram	slanted sides

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A figure can be folded along a line so both halves coincide; e.g. letter A.	1+1
12	2: the lines joining midpoints of opposite sides.	1+1
13	A, M, T (also U, V, W, Y, H, I, O, X).	1+1
14	When a figure matches itself after turning less than 360° about a fixed point.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	3; 2; infinite (3).	3
16	Fold along a line (1.5); if halves match exactly it is a line of symmetry, e.g. a heart shape (1.5).	1½+1½
17	Order 4 (1.5); angle 90° (1.5).	1½+1½
	OR: H and O (also I, X) (3).	1+2
18	Mirror image reversed left to right (1.5); distances from the line are equal (1.5).	1½+1½
	OR: 6 (1); 6 (1); 60° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Definition (1). (ii) 4; 3; 5 (2). (iii) Diagrams (2).	1+2+2
	OR: (i) Definition (1). (ii) 4, 90°; 3, 120°; 6, 60° (3). (iii) Hexagon (1).	1+3+1
20	(i) Vertical: A, H, O; horizontal: B, E, H, O; none: N, S, Z (3). (ii) H, N, S, Z, O (2).	3+2
	OR: (i) 6 (1). (ii) 6 (1). (iii) 60° (1). (iv) Yes (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 9: **Symmetry** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** line of symmetry	mirror line
2	**(d)** infinitely many	any diameter
3	**(c)** A	left and right match
4	**(b)** the mirror line	equal distances
5	**(b)** 3	fits 3 times
6	**(d)** 2	half turn
7	**(c)** rotational symmetry of order 2	half turn
8	**(d)** no line of symmetry	but has rotational symmetry
9	**(b)** MOM	symmetric
10	**(b)** the vertex between equal sides and the midpoint of the base	median

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$360^\circ \div 6 = 60^\circ$.	1+1
12	3.	1+1
13	Two diagonals and two lines joining midpoints of opposite sides (4 lines).	1+1
14	Regular pentagon.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	N 2; S 2; Z 2; X 2 (3).	3
16	Line from vertex to midpoint of base (1.5); only two sides equal (1.5).	1½+1½
17	Yes, 2 lines (1.5); along the diagonals (1.5).	1½+1½
	OR: Butterfly wings, flower petals, rangoli, Taj Mahal (3).	3
18	$360^\circ \div 3 = 120^\circ$ (3).	1+2
	OR: 8 lines (1.5); 45° (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Property (2). (ii) Correct drawing (2). (iii) Measure (1).	2+2+1
	OR: (i) Steps (2). (ii) Folds (2). (iii) 4 (1).	2+2+1
20	(i) 24 (1). (ii) 15° (1). (iii) 24 (1). (iv) It matches after every 15° (2).	1+1+1+2
	OR: (i) Rectangle 2, 2; rhombus 2, 2; parallelogram 0, 2; kite 1, 1 (3). (ii) Reasons (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.